ngladesh Army University of Engineering & Technology (BAUET)

Department of Computer Science and Engineering (CSE)

B.Sc. Engineering 1st Year 2nd Semester, Final Examination, Summer-2021

Code: EEE 1261

arks: 70

Course Title: Electrical Circuit Analysis

Time: 03 Hours

:

1. Figures shown in the right margin indicate full marks
2. Answer **Six** questions, taking **Three** from each section **including Q.1**
3. Use separate Answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Q.1 (a) Define ideal and practical voltage sources with example. 2

(b) Suppose, a bulb is connected to a circuit. Supply voltage is 200 V and bulb resistance is 20 Ω. The amount of current flow through the bulb is explained by Ohm's law. Write short note on Ohm's law and determine the amount of current flowing through the bulb. 4

(c) Consider the following circuit of fig 1.c and determine the number of nodes and loops in this circuit. Also, determine the node voltages using any circuit analysis technique. $5\frac{2}{3}$

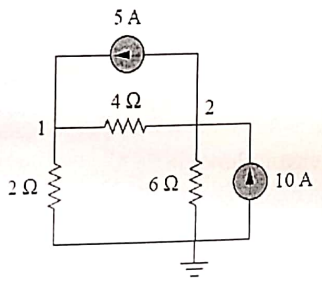

Fig. 1.c

Q.2 (a) Explain voltage divider rule in series circuit. $3\frac{2}{3}$

(b) Calculate the equivalent resistance R_{eq} in the circuit of fig 2.b. 4

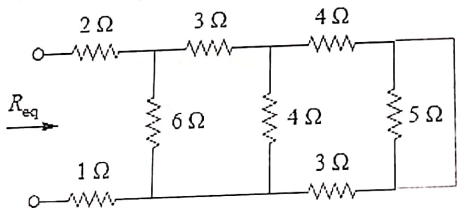

Fig. 2.b

(c) State Thevenin's theorem. Write down the steps of calculating R_{Th} for any circuit. 4

Q.3 (a) Write down the steps to determine the mesh currents using mesh analysis technique. 3

(b) Write shorts notes on maximum power transfer theorem. 4

(c) Find the value of v and i of the following circuit in fig 3.c. $4\frac{2}{3}$

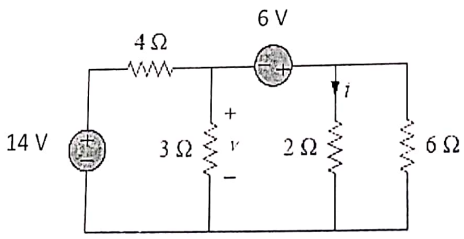

Fig. 3.c

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Q.4 (a) Define super node and super mesh with example.

(b) What do you mean by linearity? Which circuit theorem is not applicable to a non-linear circuit?

(c) Calculate the mesh currents i_1 and i_2 in the circuit shown in fig 4.c.

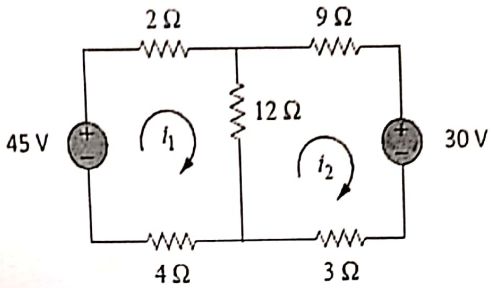

Fig. 4.c

Section-B

Q.5 (a) Define phasor and power factor. 3

(b) Find the phase angle between 4

$$i_1 = -4\sin(377t+55^\circ)$$
$$i_2 = 5\cos(377t-65^\circ)$$

Does i_1 lead or lag i_2?

(c) Find $v(t)$ and $i(t)$ in the circuit shown in fig 5.c. $4\frac{2}{3}$

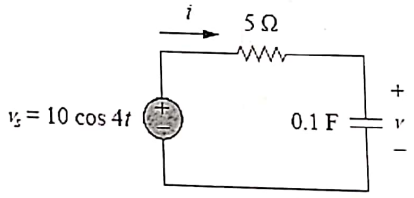

Fig. 5.c

Q.6 (a) Define instantaneous power and average power. 2

(b) Find the value of v_x in the circuit of fig 6.b using source transformation theorem. $5\frac{2}{3}$

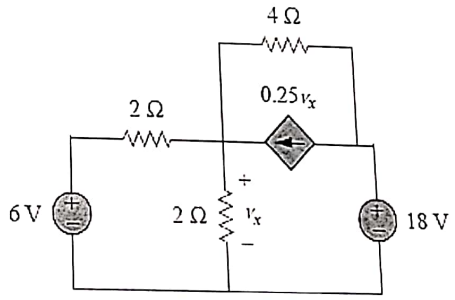

Fig. 6.b

(c) Determine the value of R_L and V_{ab} for the circuit of fig 6.c. 4

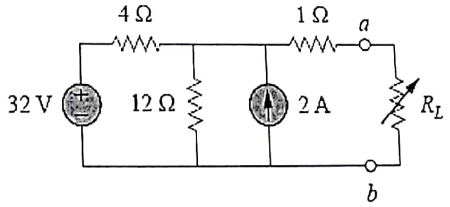

Fig. 6.c

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Find the value of Thevenin's equivalent circuit of fig 7.a. 5

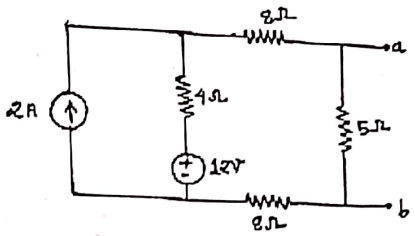

Fig. 7.a

(b) Determine $v_o(t)$ in the circuit of fig 7.b. $6\frac{2}{3}$

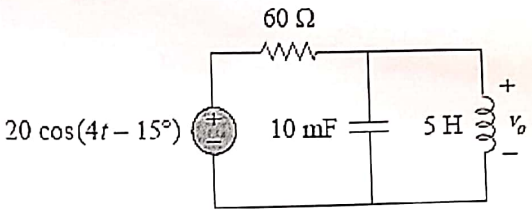

Fig. 7.b

Q.8 (a) Write down the benefits of three phase system over single phase system. 2

(b) If the current $i(t)$ of fig 8.b is passing through a 2 Ω resistor, find the average power absorbed by it. $5\frac{2}{3}$

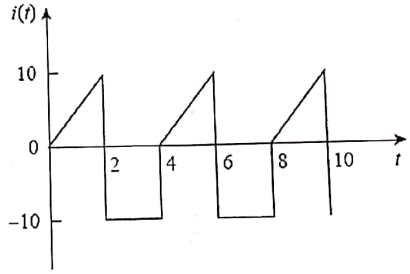

Fig. 8.b

(c) Calculate the line currents in the three phase three wire Y-Y system of fig 8.c. 4

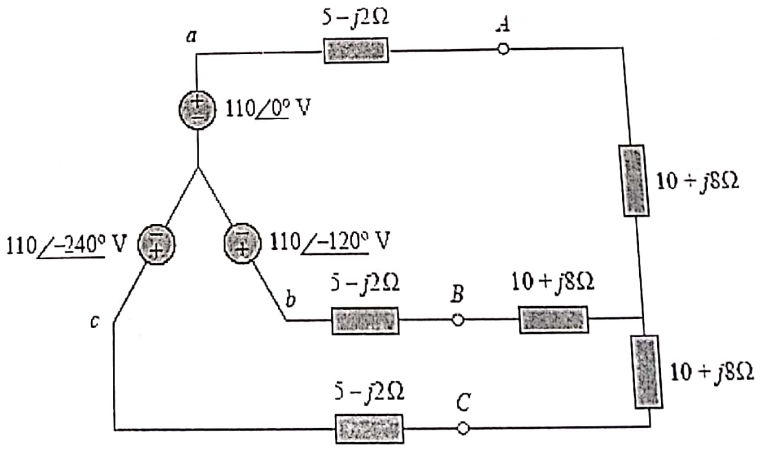

Fig. 8.c

Bangladesh Army University of Engineering & Technology (BAUET)

Department of Electrical and Electronic Engineering (EEE)

B.Sc. Engineering 1st Year 2nd Semester Referred/Improvement Examination, Fall 2022

Course Code: EEE-1261/EEE-1163

Full Marks: 70

Course Title: Electrical Circuit Analysis

Time: 3 Hours

N.B.:

1. **Figures shown in the right margin indicate full marks**
2. **Answer Six questions, taking Three from each section including Q.1 and Q.5**
3. **Use separate Answer script for each section.**
4. **Assume reasonable data if necessary.**

Section-A

		CO	PO	BL	Mark
1.	(a) State one Kirchhoff's Law that is used in Mesh analysis and also show an example with diagram.	CO1	PO1	C2	3
	(b) Explain dependent source and independent source with example.	CO1	PO1	C2	$4\frac{2}{3}$
	(c) Determine the equivalent resistance between a and b node of the following figure 1(c).	CO1	PO1	C3	4

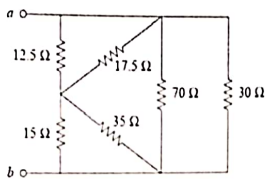

Fig.1(c)

2. (a) State and explain KVL with proper example. $2\frac{2}{3}$

(b) Find v_1 and v_2 in the circuit of Fig. 2(b) and also determine the power of 2Ω. 3

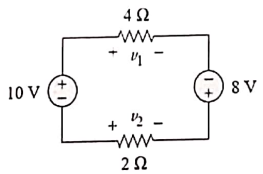

Fig. 2(b)

(c) Find voltages and currents shown in Fig. 2(c) 6

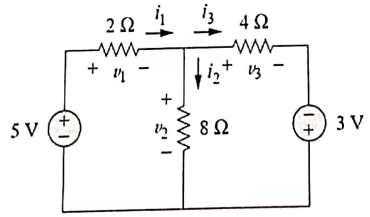

Fig. 2(c)

3. (a) State maximum power transfer theorem and derive the condition of maximum power transfer theorem. $3\frac{2}{3}$

(b) Find R_L for maximum power transfer in Fig. 3(b) and also determine that maximum power. 4

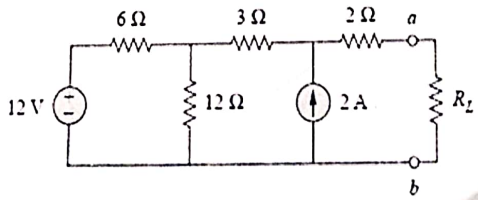

Fig. 3(b)

(c) Find v and i in the circuit of Fig. 3(c) 4

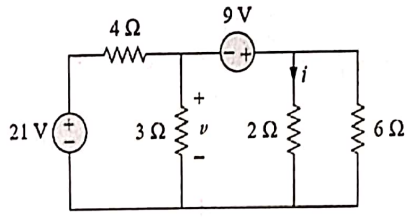

Fig. 3(c)

4. (a) Explain source transformation method with proper example. 3

(b) In Fig. 4(b) find the current through 4Ω resistance in downward direction by superposition theorem. 4

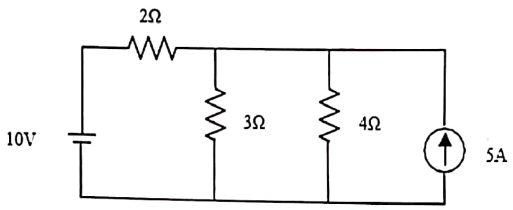

Fig. 4(b)

(c) Find Thevenin's and Norton's equivalent circuit in the terminal a and b node for the circuit in Fig. 4(c). $4\frac{2}{3}$

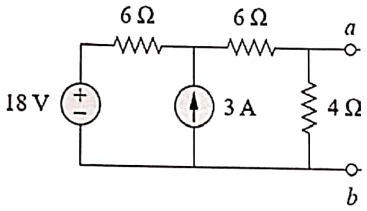

Fig. 4(c)

Section-B

		CO	PO	BL	Mark
5. (a)	Evaluate the complex number: $[(5+j2)(-1+j4)-5\angle 60^0]^*$	CO2	PO1	C2	3
(b)	Given, V(t)= 350 sin(377t+60^0) and i(t)= -10 cos(377t+30^0). Write these signal in phasor form and determine which one is lagging?	CO2	PO1	C1	$4\frac{2}{3}$
(c)	V(t)= 12 cos (50t+10^0), Find out the amplitude, phase, period and frequency of the sinusoidal signal.	CO2	PO1	C2	4

6. (a) Write down the voltage and current relationship of inductor and capacitor. 2

(b) Determine C_{eq} between terminal a-b in Fig. 6(b). 4

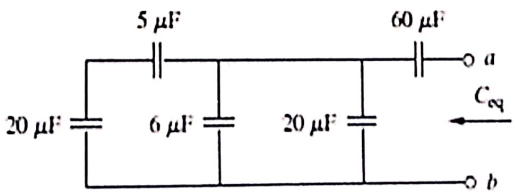

Fig. 6(b)

(c) Determine the input impedance, admittance, conductance, and susceptance of the following circuit at $\omega = 20$ rad/s in Fig. 6(c). $5\frac{2}{3}$

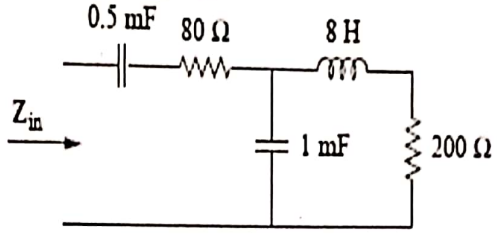

Fig. 6(c)

7. (a) 'Voltage divider rule is always applied in series circuit', Justify. 3

(b) Transform wye network to a del network in Fig 7(b). 3

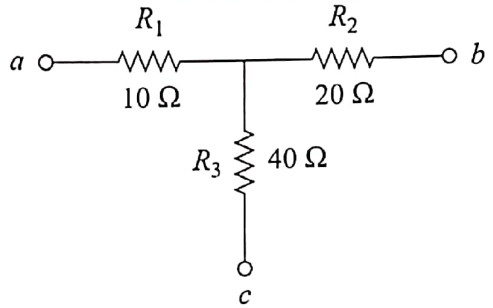

Fig 7(b)

(c) Calculate V by node and mesh analysis in Fig 7(c). $5\frac{2}{3}$

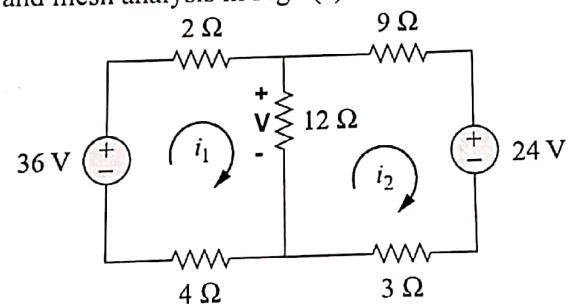

Fig 7(c)

8. (a) Discuss the advantages of 3φ system over 1φ system. 3

(b) Show that the 3φ system uses only 75% of the material those used in the equivalent 1φ system. 4

(c) Find the expression of generalized line currents of the Balanced Y-Y networks. $4\frac{2}{3}$

Bangladesh Army University of Engineering & Technology(BAUET)

Department of Computer Science and Engineering (CSE)

B.Sc. Engineering 1st Year 2nd Semester, Final Examination, Summer-2023

Course Code: EEE 1261 | Course Title: Electrical Circuit Analysis

Marks: 70 | Time: 3 Hours

Note:
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2. Answer Six questions, taking Three from each section including Q.1, Q.2 &Q.5, Q.6.
3. Use separate Answer script for each section.
4. Assume reasonable data if necessary.

Section-A

		Marks	CO	PO	KP	BL
Q.1	Imagine you are an engineer responsible for designing the electrical system of a futuristic eco-friendly city. In the heart of the city, there's a massive interconnected grid powered by a central generator, which supplies power to various districts. To ensure optimal power distribution and energy efficiency, you have a complex network consisting of 8 resistors (R1 to R8) placed strategically throughout the city's power grid (Fig.1).					

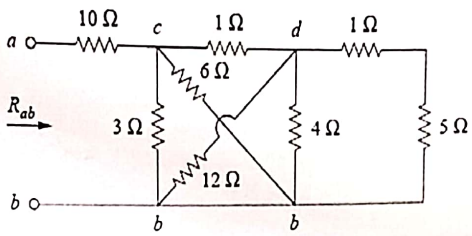


Fig.1

	(a) Determine the total equivalent resistance of this network.	04	CO1	PO2	-	C2
	(b) In the context of the circuit depicted in Fig.1, connected to a 60V DC source, determine the total power supplied by the central generator to the entire city, while considering the overall equivalent resistance; also find out the power factor.	05	CO1	PO2	-	C3
	(c) Identify the voltage drop and current flowing through a specific resistor (3Ω) to optimize power distribution in a critical part of the city.	03	CO1	PO3	-	C3
Q.2	(a) Discuss the condition of series and parallel of electrical circuit elements. Simulation often arise in circuit analysis when the loads are neither in parallel nor in series. In this circumstance what kind of technique will you use to solve the circuit and how?	04	CO1	PO1	-	C2
	(b) Explain why node and mesh analysis may not be suitable for all electrical circuits and also approaches to overcome this limitations.	03	CO1	PO1	-	C3
	(c) Determine the current through 7Ω resistor by supernode or super mesh in Fig. 2(c).	05	CO2	PO2	-	C3

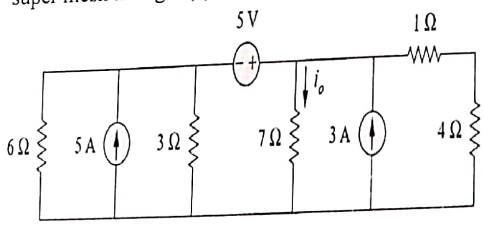
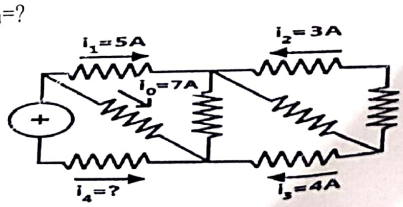


Fig. 2(c)

Q.3	(a) Explain the significance of Wye-Delta Transformation in an electrical Circuit.	02				
	(b) Identify the current, i_4=?	04				



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(c) Find the current supplied by the d.c. source in the circuit shown in Fig. 3(c). 05

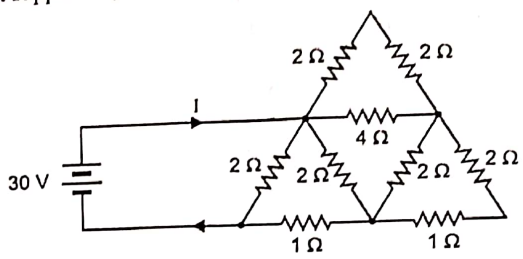

Fig.3(c)

Q.4 (a) State maximum power transfer theorem for DC circuit and maximum average power transfer theorem for AC circuit. 03

(b) Find I in the circuit in fig.4(b) 04

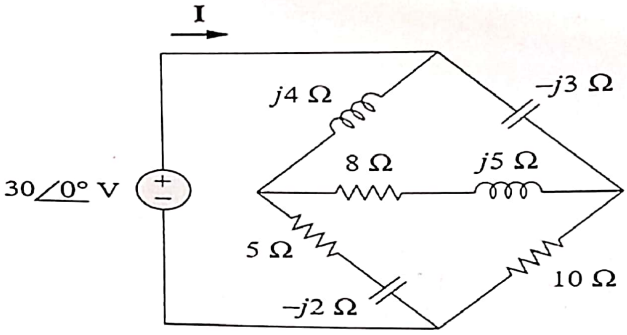

Fig.4(b)

(c) Find the value of R_L for maximum power transfer in the circuit given in Fig. 4(c) and also determine that maximum power. 04

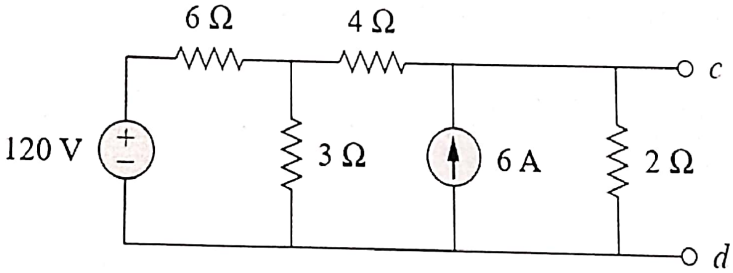

Fig. 4(c)

Section-B

Q.5 Imagine you are an electrical engineer working at a music concert venue. The stage is equipped with a complex sound system that uses a series-parallel ladder network of speakers, amplifiers, and filters.

The speakers and various audio components within the ladder network have different impedance values. The concert is about to start, and you need to ensure that the entire sound system operates efficiently.

Marks CO PO KP BL

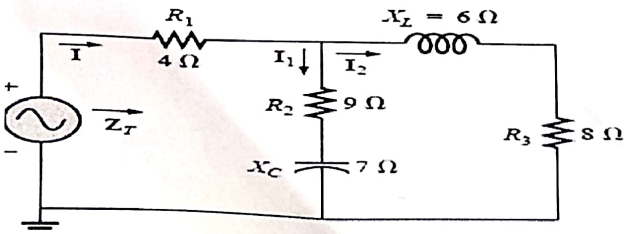

Fig. 5

Q.5	(a)	Explain RL, RC, RLC Circuits.	03			
	(b)	Identify the total impedance of the ladder network (Fig.5) to understand how the sound system responds to different audio frequencies.	03	CO3	-	-	C2
	(c)	Determine the total current flowing through the network (Fig.5) to ensure that the amplifiers can provide sufficient power to the speakers, if the supply voltage is $100\angle 0^0$ V AC.	03	CO3	PO2	-	C3
	(d)	Find the average power delivered to the network (Fig.5).	03	CO3	PO2	-	C3
Q.6	(a)	Discuss the source transformation theorem with it's statement condition and implementation.	03	CO2	PO2	-	C2
	(b)	Calculate V_0 in the circuit in Fig. 6(b)	05	CO2	PO2	-	C3

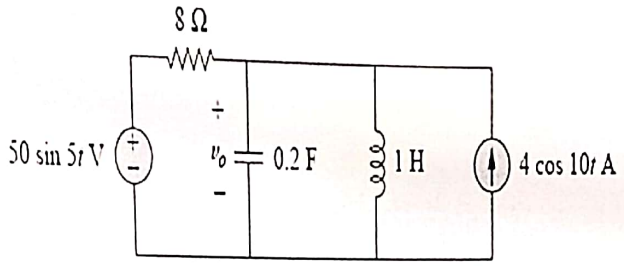

Fig. 6(b)

	(c)	Given two AC voltages Waveforms, $V_1 = 10\cos(211t - 30^0)$ $V_2 = 22\sin(211t + 40^0)$ Calculate the phase difference between them and also determine which one leads the other.	04	-	-	-	C3
Q.7	(a)	Discuss the advantages of 3φ system over 1φ system.	03				
	(b)	Show that the three-phase system uses only 75 percent of the material those used in the equivalent single-phase system.	04				
	(c)	Consider for single phase system phase voltage v(t)=120 cos (100t-20)and phase current is and i(t)= 4 cos (100t+10). Find (i) Complex power, (ii) Apparent power, (iii) Real power and Reactive power, (iv) P.F and (vi) Load impedance.	04				
Q.8	(a)	Explain the conditions of resonance for series circuit.	03				
	(b)	In the Fig.8(b) RLC Circuit, R=1000Ω, resonant frequency=159KHz, C=10μF. Identify the value of L under resonant condition. Also, find the value Q (Quality Factor) and Bandwidth(B.W).	04				

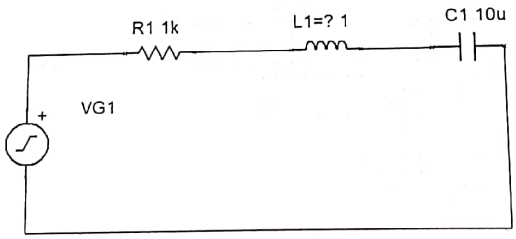

Fig.8(b)

	(c)	Using Kirchhoff's voltage law, determine the circuit current in Fig.8(c)	04			

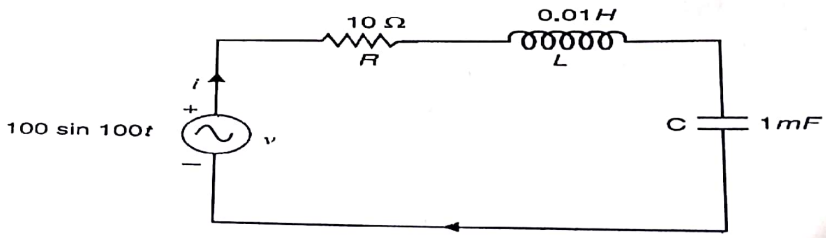

Fig.8(c)

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...ion, Summer-2022

Course Code: EEE-1261

Full Marks: 70

Course Title: Electrical Circuit Analysis

Time: 3 Hours

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2. Answer Six questions, taking Three from each section including Q.1
3. Use separate Answer script for each section.
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Section-A

Q.1 (a) 'In DC circuit, the behavior of an inductor is like an open circuit', justify your answer. 3

(b) State one Kirchhoff's law with example that is used in mesh analysis. 2

(c) Determine R_{eq}, V and the current through 35Ω for the fig. 1(c). $6\frac{2}{3}$

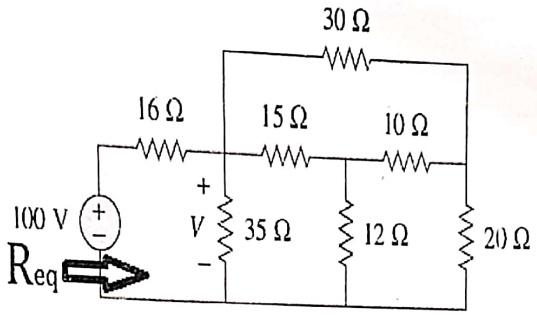

Fig.1(c)

Q.2 (a) Define node and loop of a circuit. $1\frac{2}{3}$

(b) Determine the amplitude, phase, period and frequency of the sinusoid, $V(t)= 12 \cos(50t+10^0)$. 4

(c) Find currents and voltages for fig.2(c) 6

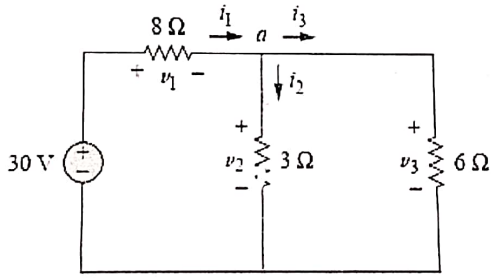

Fig. 2(c)

Q.3 (a) State and differentiate between maximum power transfer theorem and maximum average power transfer theorem. 3

(b) Derive the condition of maximum power transfer theorem and from this also prove that $P_{max}=V_{th}^2/(4 R_{th})$. $4\frac{2}{3}$

(c) the value of R_L for maximum power transfer in the circuit given in fig. 3(c) and also determine that power. 4

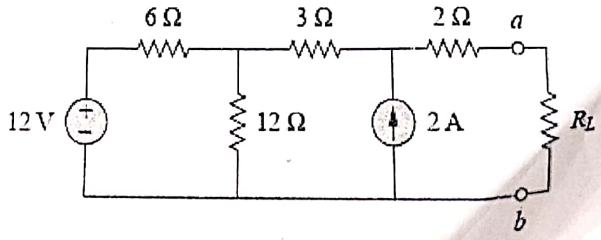

Fig. 3(c)

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Q.4 (a) Find the reason for using superposition theorem. $1\frac{2}{3}$

(b) Determine the current, I through 3Ω from the given direction in fig. 4(b) using superposition theorem. 5

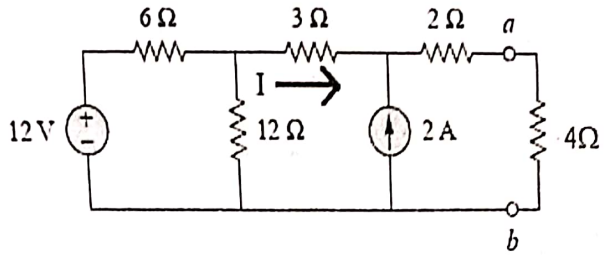

Fig. 4(b)

(c) Determine the equivalent inductance of the circuit shown in fig.4(c) 3

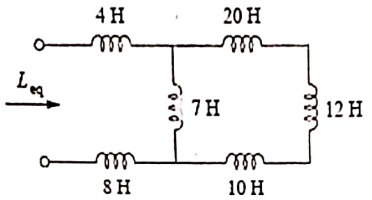

Fig. 4(c)

(d) If voltage v=6 Sin(100t−30⁰) is applied to a 50μF capacitor, calculate the current through the capacitor. 2

Section-B

Q.5 (a) State the theorem that the current source can be converted by a voltage source. $1\frac{2}{3}$

(b) Find Thevenin's and Norton's equivalent circuit between terminal a and b in fig.5(b). 6

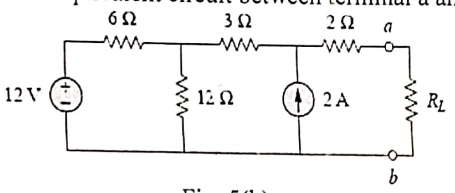

Fig. 5(b)

(c) Use source transformation to find V_0 in the circuit of fig. 5(c) 4

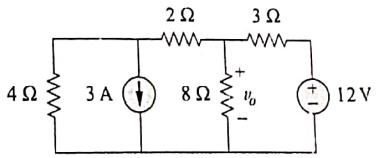

Fig. 5(c)

Q.6 (a) Find the ***phase angle*** between $V_1 = -4\sin(37t + 25^0)$ and $V_2 = -5\cos(37t + 40^0)$. And also determine which one is lead? $2\frac{2}{3}$

(b) For the circuit in fig.6(b) find Z_{in}, Y_{in} & B_{in} and also find the voltage $V_0(t)$. 6

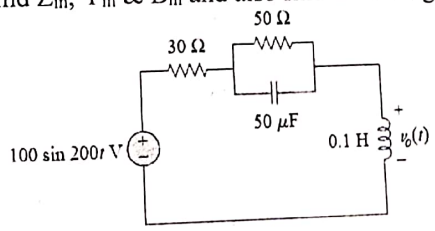

Fig. 6(b)

(c) Find the wattmeter reading of the circuit in fig. 6(c) 3

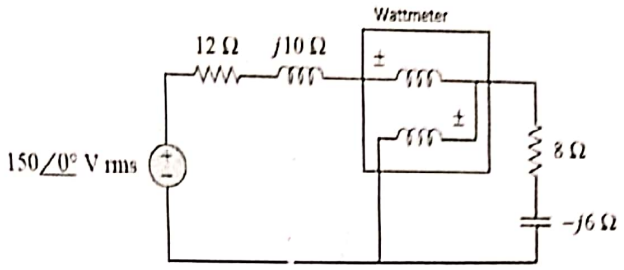

Fig.6(c)

Q.7 (a) Given, $V_{bn}=120\angle 150^0$, find V_{an} & V_{cn} for positive sequence. 2

(b) Given that, $V(t)= 120 \cos(377t+45^0)$ and $i(t)= 10 \cos(377t-10^0)$. Find the instantaneous power and average power. 3

(c) Show that the three-phase system uses only 75% of the material those used in the equivalent single-phase system. 3

(d) Find the expression of the line currents and phase currents in the balanced Δ-Δ three-phase circuit. $3\frac{2}{3}$

Q.8 (a) Define self and mutual inductance. 2

(b) Calculate the total inductance L_{AB}, for this magnetically coupled circuit in fig. 8(b). 3

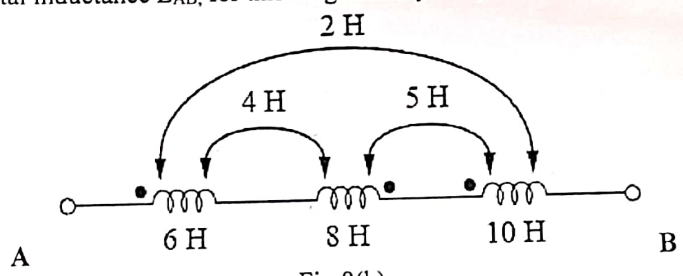

Fig.8(b)

(c) Analyze the circuit shown in fig. 8(c) to determine the branch currents I_1, I_2 and I_3 using mesh analysis. $6\frac{2}{3}$

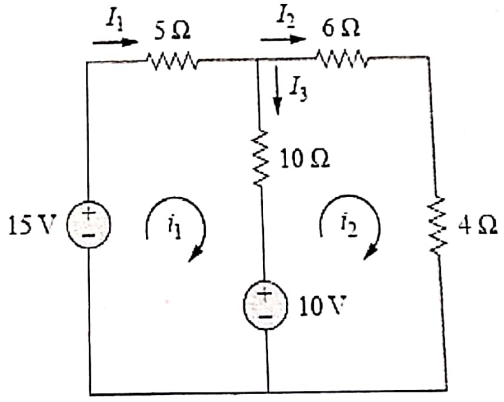

Fig. 8(c)

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